1) Lesson 9/10/26

1.1) Inequalities

General Procedure:

1. Determine the domain D.
2. Bring all terms to one side: $f(x) \gtrless 0$.
3. Factor into products or fractions: $\frac{A(x)}{B(x)} \gtrless 0$.
    1. Find the roots of each factor.
    2. Construct a sign chart.
4. Intersect the resulting intervals with the domain D.

There can happen 3 main cases: fractions, square roots or absolute values

Ex c) Fraction

Solve the following inequality:

$$\frac{x-1}{x-2} > \frac{2x-3}{x-3}$$

Domain:

$$D = x \in \mathbb{R} \setminus \{2, 3\}$$

Simplify:

$$\frac{x-1}{x-2} - \frac{2x-3}{x-3} > 0 \implies \frac{(x-1)(x-3) - (2x-3)(x-2)}{(x-2)(x-3)} > 0$$

This is of form $A(x)/B(x)$ so the study of the numerator and denominator needs to be done independently

Numerator:
Find zeroes:

$$\begin{aligned}(x-1)(x-3) - (2x-3)(x-2) &= (x^2 - 4x + 3) - (2x^2 - 7x + 6) \\ &= -x^2 + 3x - 3\end{aligned}$$

Recall the quadratic formula:
Let $a, b, c \in \mathbb{R}$, then $ax^2 + bx + c = 0$ has the following solution:

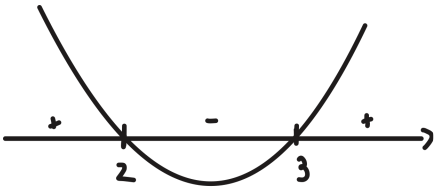
$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Where the discriminant $\Delta = b^2 - 4ac$ determines the types of solution:

- $\Delta > 0$: two real (distinct) solutions
- $\Delta = 0$: two real (identical) solutions
- $\Delta < 0$: no real solution

In this case since $\Delta = 9 - 12 = -3 < 0$ we don't have any solutions, hence since $a < 0$ the whole numerator is negative.

Denominator:
Here the function already shows the zeroes, so we can plot the graph to see the regions:



Curve

So we have positive denominator for $x < 2 \cup x > 3$.

By noticing the interval where they have the same sign:

x	$(-\infty, 2)$	$(2, 3)$	$(3, +\infty)$
Numerator	$-$	$-$	$-$
Denominator	$+$	$-$	$+$
Fraction	$-$	$+$	$-$

Finally we have:

$$\boxed{2 < x < 3}$$

Ex d) Absolute value

Solve the following inequality:

$$\frac{|x|}{x-1} > \frac{x+1}{2x-1}$$

Domain:

$$D = x \in \mathbb{R} \setminus \left\{ \frac{1}{2}, 1 \right\}$$

Now recall the abs value definition:

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

In this case $f(x) = x$ so we have:

$$\begin{cases} \dfrac{x}{x-1} > \dfrac{x+1}{2x-1} & x \geq 0 \\ \dfrac{-x}{x-1} > \dfrac{x+1}{2x-1} & x < 0 \end{cases}$$

Remeber:

- The solution to each part of the system must be put in intersection with the condition
- The solutions of the parts of a system are in union with each other

Start with case 1:
Rewrite so that it is a single fraction:

$$\frac{x}{x-1} - \frac{x+1}{2x-1} > 0 \implies \frac{x(2x-1) - (x+1)(x-1)}{(x-1)(2x-1)} > 0$$

- Numerator:

$$x(2x-1)-(x^2-1)=2x^2-x-x^2+1=x^2-x+1$$

Find the zeroes of the numerator:

$$x^2-x+1=0 \to \Delta=-3$$

Since $\Delta<0$ and $a=1>0$, the numerator is strictly positive.

- Denominator:

$$(x-1)(2x-1)>0 \iff x<\frac{1}{2} \cup x>1$$

Sign analysis for Case 1:

x	$\left(-\infty,\frac{1}{2}\right)$	$\left(\frac{1}{2},1\right)$	$(1,+\infty)$
Numerator	$+$	$+$	$+$
Denominator	$+$	$-$	$+$
Fraction	$+$	$-$	$+$

So the fraction is >0 on $\left(-\infty,\frac{1}{2}\right)\cup(1,+\infty)$.

Intersecting with the case condition $x\geq 0$:

$$S_1=\left[0,\frac{1}{2}\right)\cup(1,+\infty)$$

Case 2:
Same procedure as case 1:

$$\frac{-x}{x-1}-\frac{x+1}{2x-1}>0 \implies \frac{-x(2x-1)-(x+1)(x-1)}{(x-1)(2x-1)}>0$$

- Numerator:

$$-2x^2+x-(x^2-1)=-3x^2+x+1$$

Find the zeroes of the numerator:

$$x_{1,2}=\frac{-1\pm\sqrt{13}}{-6}=\frac{1\mp\sqrt{13}}{6}$$

Since $a=-3<0$, the parabola opens downward:

$$N>0 \iff x_1=\frac{1-\sqrt{13}}{6}<x<\frac{1+\sqrt{13}}{6}=x_2$$

Denominator:
The denominator roots are $\frac{1}{2}=0.5$ and 1.

Sign analysis for the factors:

x	$(-\infty,x_1)$	$(x_1,1/2)$	$(1/2,x_2)$	$(x_2,1)$	$(1,+\infty)$
Numerator	$-$	$+$	$+$	$-$	$-$
Denominator	$+$	$+$	$-$	$-$	$+$
Fraction	$-$	$+$	$-$	$+$	$-$

The fraction is > 0 on $\left(x_1, \frac{1}{2}\right) \cup (x_2, 1)$.

Intersecting with the case condition $x < 0$:

$$S_2 = \left(\frac{1-\sqrt{13}}{6}, 0\right)$$

Taking the union of both cases $S = S_1 \cup S_2$:

$$S = \left(\frac{1-\sqrt{13}}{6}, 0\right) \cup \left[0, \frac{1}{2}\right) \cup (1, +\infty)$$

Notice that 0 is included in the solution set, so the first two intervals join together:

$$\boxed{S = \left(\frac{1-\sqrt{13}}{6}, \frac{1}{2}\right) \cup (1, +\infty)}$$

Ex f) Square Root

Solve the following inequality:

$$\sqrt{x^2 - 6x} > x + 2$$

First notice that the domain is

$$x^2 - 6x \geq 0 \to D = (-\infty, 0] \cup [6, +\infty)$$

First notice that when $x + 2 < 0$ the solution inequality holds (in the domain). So one part of the solution is:

$$S_1 = D \cap \{x < -2\} = \{x < -2\}$$

Now for $x + 2 \geq 0$ of both sides can be squared:

$$x^2 - 6x > x^2 + 4x + 4 \to 0 > 10x + 4 \to x < -\frac{2}{5}$$

This clearly for the values bigger than -2 in the domain, that is

$$S_2 = D \cap \left\{\{x \geq -2\} \cap \left\{x < -\frac{2}{5}\right\}\right\} = D \cap \left\{-2 \leq x < -\frac{2}{5}\right\} = \left\{-2 \leq x < -\frac{2}{5}\right\}$$

So the result is:

$$S = S_1 \cup S_2 = \{x < -2\} \cup \left\{-2 \leq x < -\frac{2}{5}\right\} = x < -\frac{2}{5}$$

Ex i) Mixed Together

Solve the following inequality:

$$\sqrt{|x^2 - 4|} - x \geq 0$$

Domain: $|x^2 - 4| \geq 0 \to D = \mathbb{R}$

Now recall the abs value definition:

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

In this case $f(x) = x^2 - 4 = (x+2)(x-2)$ so $f(x) \geq 0 \rightarrow x \leq -2 \cup x \geq 2$

Now we can rewrite the inequality as a system of two inequalities:

$$\sqrt{|x^2-4|} - x \geq 0 \rightarrow \begin{cases} \sqrt{x^2-4} - x \geq 0 & x \leq -2 \cup x \geq 2 \\ \sqrt{4-x^2} - x \geq 0 & -2 < x < 2 \end{cases}$$

Each system can be solved independently from the other.
Call the conditions of the cases D_1 and D_2 respectively.

Solve first case:
Recall that when RHS (x) is negative there is always a solution.

$$\sqrt{x^2-4} \geq x$$

Therefore one part of the solution includes $x < 0$

Now study in the interval $x \geq 0$: since we are sure that both sides are ≥ 0 we can square both sides:

$$x^2 - 4 \geq x^2 \rightarrow -4 \geq 0$$

Not possible, so the only solution to this part is when x is negative:

$$S_1 = x \leq 0 \cap D_1 = x \leq -2$$

Solve the second equation:
As before a solution always exists for $x < 0$. Now consider only the positive values ($x \geq 0$) in order to be able to square both sides:

$$\sqrt{4-x^2} \geq x \rightarrow 4 - x^2 \geq x^2 \rightarrow 4 - 2x^2 \geq 0 \rightarrow -\sqrt{2} \leq x \leq \sqrt{2}$$

So we have $S_2 = D_2 \cap \left\{ x < 0 \cup -\sqrt{2} \leq x \leq \sqrt{2} \right\} = D_2 \cap x \leq \sqrt{2} = -2 < x \leq \sqrt{2}$

The result is therefore the union of both cases which yields:

$$\boxed{S = x \leq -2 \cup -2 < x \leq \sqrt{2} \rightarrow S = x \leq \sqrt{2}}$$

1.2) Subsets of $\mathbb{R}$

Ex a)

Describe the following set:

$$A = S_1 \cap S_2 = \left\{ x \in \mathbb{R} : x^2 + 4x + 13 < 0 \right\} \cap \left\{ x \in \mathbb{R} : 3x^2 + 5 > 0 \right\}$$

Subset 1:

$$x^2 + 4x + 13 < 0 \rightarrow \Delta = -36 \implies S_1 = \emptyset$$

any intersection with an empty set is an empty set:

$$\boxed{A = \emptyset \ \cap \ S_2 = \emptyset}$$

In general recall:

$$\emptyset \cap S = \emptyset \qquad \emptyset \cup S = S$$

Ex c)

Describe the following set:

$$C = \left\{ x \in \mathbb{R} : \frac{x^2 - 5x + 4}{x^2 - 9} < 0 \right\} \cup \left\{ x \in \mathbb{R} : x + \sqrt{7x + 1} = 17 \right\}$$

Subset 1:
First notice that the domain is $x \neq \pm 3$

Numerator:

$$x^2 - 5x + 4 = 0 \to x_{1,2} = 1,\ 4 \to N > 0 \iff x < 1 \cup x > 4$$

Denominator:

$$x^2 - 9 = (x + 3)(x - 3) = 0 \to D > 0 \iff x < -3 \cup x > 3$$

x	$(-\infty, -3)$	$(-3, 1)$	$(1, 3)$	$(3, 4)$	$(4, +\infty)$
Numerator	$+$	$+$	$-$	$-$	$+$
Denominator	$+$	$-$	$-$	$+$	$+$
Fraction	$+$	$-$	$+$	$-$	$+$

so $S_1 = (-3, 1) \cup (3, 4)$

Subset 2:
This is just an equality:

$$\sqrt{7x + 1} = 17 - x$$

The domain is $7x + 1 \geq 0 \to D = \left\{ x \geq -\frac{1}{7} \right\}$
Moreover as before, both sides must have the same sign when squaring and since the sqrt is always ≥ 0 we must have that also $17 - x \geq 0 \to x \leq 17$ so we can further reduce the domain to

$$D = \left[-\frac{1}{7}, 17 \right]$$

Square both sides:

$$7x + 1 = (17 - x)^2 \to 7x + 1 = x^2 - 34x + 17^2 \to x^2 - 41x + 288 = 0$$

$$x_{1,2} = 9,\ 32$$

But only 9 is acceptable as $32 \notin D$

Finally:

$$\boxed{C = (-3, 1) \cup (3, 4) \cup \{9\}}$$

1.3) Max/Min, Sup/Inf

Recall the theory

Definition (Upper/Lower Bound)**.**

$M \in \mathbb{Q}$ is called **upper bound** for A if $M \geq a, \forall a \in A$.

Respectively it is called **lower bound** if $M \leq a, \forall a \in A$

Notice M does not need to be in A
Example: consider set $A = [0, 1)$ any $m \leq 0$ is a lower bound and any $M \geq 1$ is upper bound

Definition (Maximum/Minimum of A)**.**

If $\overline{M}$ is an upper bound for A and $\overline{M} \in A$, then $\overline{M}$ is the *minimum* possible *upper* bound and is the **maximum** (of A).

If $\overline{m}$ is an lower bound for A and $\overline{m} \in A$, then $\overline{m}$ is the *maximum* possible *lower* bound and the **minimum** (of A).

In this case $\overline{M} \in A$ by definition.
From the previous example: $\overline{m} = \max\{m \leq 0\} = 0 \in A$ is the correct minimum of A, however $\overline{M} = \min\{M \geq 1\} = 1 \notin A$ means that we don't have a maximum

Definition (Supremum/Infimum)**.**

The **supremum of** A $\sup A$ is the minimum of the upper bounds of A.

The **infimum of** A $\inf A$ is the maximum of the lower bounds of A.

Finish the example: we know $\overline{m} = 0 \in A$ so this is both minimum and infimum of A. We know also $\overline{M} = 1 \notin A$, therefore $\sup \mathrm{A} = 1$ but it is not the maximum.

Recall that if a set is unbounded, then

$$\sup A = +\infty \quad \text{if } A \text{ is unbounded from above}$$
$$\inf A = -\infty \quad \text{if } A \text{ is unbounded from below}$$

Ex a)

Tell wether this subsets is bounded and the sup/inf and if it allow for maximum/minimum.

$$A = \left\{ x \in \mathbb{R} : x = n \text{ or } x = \frac{1}{n^2}, n \in N \setminus \{0\} \right\}$$

It is immediately clear that A is the union of two sets, mainly

$$A = S_1 \cup S_2 \text{ where } \begin{array}{l} S_1 = \{n \in \mathbb{N} \setminus \{0\}\} = \{1, 2, 3, 4, \ldots\} \\ S_2 = \left\{ \frac{1}{n^2} \mid n \in \mathbb{N} \setminus \{0\} \right\} = \left\{ 1, \frac{1}{4}, \frac{1}{9}, \frac{1}{16}, \ldots \right\} \end{array}$$

Find the bounds on both sets individually (not required but the result is clearer).

Study the first subset S_1: notice that it is just the natural numbers with zero excluded, so its properties are already known.

- No upper bound, so also no max
  - Supremum: since unbounded; $\sup S_1 = +\infty$
- Lower bound: $m_1 \leq 1$
  - Minimum: $\overline{m}_1 = \max m_1 \leq 1 = 1 \in S$

Study the second subset S_2: Intuitively we know that the function describing the set is monotonically decreasing: we can express it by saying:

Call x_n the value of x for any chosen n, then $x_{n+1} < x_n$.
This is easily shown as the following inequality is true for $n \in \mathbb{N} \setminus \{0\}$ (check at home):

$$\frac{1}{(n+1)^2} < \frac{1}{n^2}$$

Therefore for every number $x_n > 0$ we can find a smaller number such that $0 < x_{n+1} < x_n$

Is 0 the maximum lower bound?
By contradiction suppose there exists a lower bound $\delta > 0$ then if 0 is the maximum lower bound we have

$$\exists n \in \mathbb{N} \setminus \{0\} \quad \text{such that} \quad \frac{1}{n^2} < \delta$$

Now compute

$$\frac{1}{n^2} < \delta \to n^2 > \frac{1}{\delta} \to n > \frac{1}{\sqrt{\delta}}$$

Because $\delta > 0$, the number $\frac{1}{\sqrt{\delta}}$ is just a fixed positive real number and an appropriate n exists.

With this digression aside, we also know that $0 \notin S_2$.
As before analyze:

- Lower bound: $m_2 \leq 0$
  - No minimum: $\overline{m}_2 = \max m_2 \leq 0 = 0 \notin S_2$
  - Infimum is $\inf S_2 = 0$
- Upper bound: $M_2 \geq 1$
  - $\overline{M}_2 = \min M_2 \geq 1 = 1 \in S_2$

Putting all together:
Now consider the union, so:

- No upper bound since S_1 is unbounded
  - therefore $\sup A = +\infty$
- Lower bound: The infimum is the **maximum** of all lower bounds: $\inf A = \min(\inf S_1, \inf S_2) = \min(1, 0) = 0$. Since $0 \notin A$, $\min A$ does not exist.

2) Lesson 16/10/26

2.1) Domains

Recall:

$$\begin{aligned} \frac{a(x)}{f(x)} \quad & \text{dom } f = \{f(x) \neq 0\} \\ \sqrt{f(x)} \quad & \text{dom } f = \{f(x) \geq 0\} \\ \log(f(x)) \quad & \text{dom } f = \{f(x) > 0\} \end{aligned}$$

a) Fraction and Sqrt

Determine the domains of the following function:

$$f(x) = \frac{\sqrt{x^2 - 3x - 4}}{x + 5}$$

Here the domain is given by two solutions that must hold simultaneously.

Fraction:
Here $D_1 = \{x + 5 \neq 0\} = \{x \neq -5\}$

Sqrt:
Here $D_2 = \{x^2 - 3x - 4 \geq 0\}$. The zeroes of thsi function are $x_{1,2} = -1, 4$ so the inequality holds for $D_2 = \{x \leq -1 \cup x \geq 4\}$

Finally

$$\text{dom } f = D_1 \cap D_2 = \{x \leq -1 \cup x \geq 4\} \setminus \{-5\} = (-\infty, -5) \cup (-5, -1] \cup [4, +\infty)$$

where the last part is the correct formal way to write the result.

b) Logarithm

Determine the domains of the following function:

$$f(x) = \log(x^2 - x)$$

The domain is given by $x^2 - x > 0$, so solve the inequality:

$$x^2 - x = x(x - 1) > 0$$

The first term is positive for $x > 0$, the second for $x > 1$, so

$$\text{dom } f = \{x < 0 \cup x > 1\}$$

c) System

Determine the domains of the following function:

$$f(x) = \begin{cases} \dfrac{1}{2x + 1} & x \geq 0 \\ e^{\sqrt{x+1}} & x < 0 \end{cases}$$

Here the domain is given by the union of the two cases.

Case 1:
Here the domain is given by $D_1^* = \{2x + 1 \neq 0\} = \left\{x \neq -\frac{1}{2}\right\}$. However since this is a system, this must hold only inside the specified case, so $D_1 = D_1^* \cap \{x \geq 0\} = \{x \geq 0\}$ since $-\frac{1}{2} \notin \{x \geq 0\}$, so this part of the map is always valid.

Case 2:
Here the domain is given by $x + 1 \geq 0 \rightarrow x \geq -1$ which combined with the system is $x \in [-1, 0)$

The final domain becomes the union of the two cases, so

$$\text{dom } f = \{x \geq 0\} \cup \{-1 \leq x < 0\} = \{x \geq -1\}$$

2.2) Symmetries and Periodicity

If a function is symmetric, it can have either even or odd symmetry:

- Even Symmetry: mirrored along y axis: $f(x) = f(-x)$
- Odd Symmetry: mirrored along x and y axis: $f(x) = -f(-x)$

a) Sqrt and Abs

Sketch the graph of the following function indicating the possible symmetries and/or periodicity:

$$f(x) = \sqrt{1 - |x|}$$

Domain:

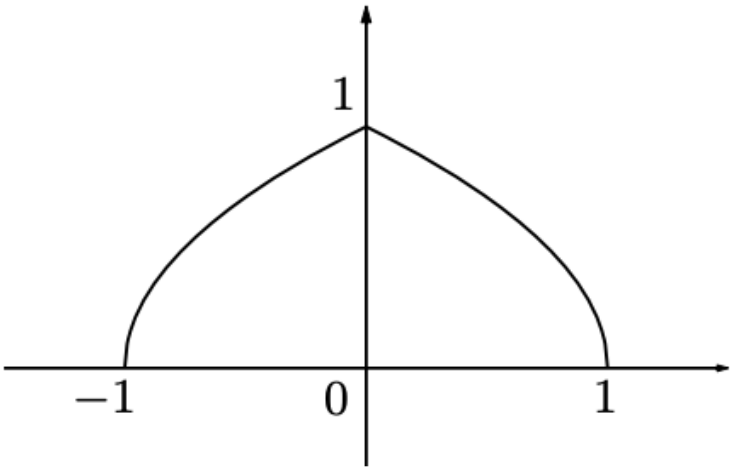
$$D = \{1 - |x| \geq 0\} = \{|x| \leq 1\} = \{x \in [-1, 1]\}$$

So clearly the function cannot be periodic. By definition, a periodic function requires $x \in \text{dom } f \implies x + T \in \text{dom } f$ for all x, which cannot hold on a bounded interval

Symmetry:
Since f is the result of a sqrt, it can only be positive (roots at $x = \pm 1$), therefore an odd symmetry is not possible (only possible if f is constant 0). Try the even symmetry:

$$f(-x) = \sqrt{1 - |-x|} = \sqrt{1 - |-1| \cdot |x|} = \sqrt{1 - 1 \cdot |x|} = f(x)$$

The final sketch can be done by sampling points only between $[0, 1]$ since then it can be mirrored.



Sketch

b) Cosine

Sketch the graph of the following function indicating the possible symmetries and/or periodicity:

$$f(x) = 1 + \cos(2x)$$

There is no restriction in the domain, so start with the symmetries. Again since $\cos(z) \in [-1, 1]$ the function returns only positive values, so an odd symmetry is excluded, moreover since $\cos(z)$ is an even function, we can guess that the function is even, let's prove it:

$$f(-x) = 1 + \cos(-2x) = 1 + \cos(2x) = f(x)$$

Periodicity:
T_1: 1 is a constant function. A constant function $c(x) = c$ satisfies $c(x+T) = c(x)$ for **every** $T > 0$. Therefore, every positive real number is a period. However it does not have a fundamental period (0 cannot be period)
T_2: $\cos(\theta)$ has period 2π, that is $\cos(\theta + 2\pi) = \cos(\theta)$, but what about $g(x) = \cos(2x)$? Let's try the generic case $\cos(ax)$, $a > 0$

By the definition of period, we must find a $T > 0$ such that

$$\min_T \{g(x+T) = g(x)\} = \min_T \{\cos(a(x+T)) = \cos(ax)\}$$

Recall that $\cos(\theta + 2\pi) = \cos(\theta)$ and by comparing it with the argument of the minimum we get

$$aT = 2\pi \rightarrow T = \frac{2\pi}{a}$$

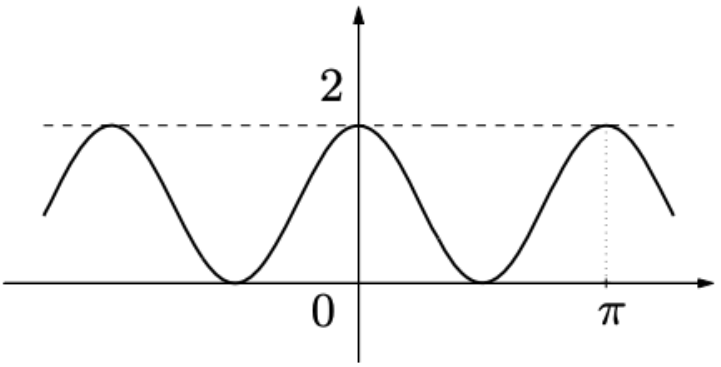
And in our case $T_2 = \pi$

Adding a constant is merely a vertical translation and leaves the periodic properties unchanged:

$$f(x+T) = f(x) \iff 1 + \cos(2(x+T)) = 1 + \cos(2x) \iff \cos(2(x+T)) = \cos(2x)$$

Therefore, the fundamental period of $f(x)$ is simply the fundamental period of $\cos(2x)$:

$$T = \frac{2\pi}{2} = \pi$$



Sketch

Two sinusoids

Determine the period of the following function:

$$f(x) = \sin(2x) + \cos(\pi x)$$

From the previous exercise we know that a sinusoidal has period

$$T = \frac{2\pi}{a}$$

So we know the two periods:

$$T_1 = \frac{2\pi}{2} = \pi \qquad T_2 = \frac{2\pi}{\pi} = 2$$

Now for the final period, recall that **the sum of two periodic functions is periodic if and only if the ratio of their periods is a rational number.**
Check the ratio:

$$\frac{T_1}{T_2} = \frac{\pi}{2} \notin \mathbb{Q}$$

so the function is aperiodic.

Now consider

$$f(x) = \sin(2x) + \cos(3x)$$

therefore

$$\begin{aligned} T_1 &= \pi \\ T_2 &= \frac{2\pi}{3} \end{aligned} \qquad \frac{T_1}{T_2} = \frac{3}{2} \in \mathbb{Q}$$

We know the function is periodic! But how is the period found?
To find the fundamental period T, any common period must satisfy $f(x+T) = f(x)$ for all x. For the sum of two independent sinusoids, this requires T to be a simultaneous integer multiple of both individual periods T_1 and T_2:

$$T = k_1 T_1 = k_2 T_2 \quad \text{for some } k_1, k_2 \in \mathbb{Z}^+$$

That is:

$$k_1\pi = k_2\left(\frac{2\pi}{3}\right) \implies k_1 = \frac{2}{3}k_2 \implies \frac{k_1}{k_2} = \frac{2}{3} \qquad k_1 = 2, k_2 = 3$$

and thus the period is

$$T = 2\pi = 3 \cdot \frac{2\pi}{3}$$

1) Arcsin

Determine domain, sign and possible symmetries and periodicity of the following function:

$$f(x) = \arcsin\left(\left|x^3 - \frac{1}{2}\right|\right)$$

Properties of $\arcsin(t)$
The sin function is not bijective (surjective but not injective) the map is $\mathbb{R} \to [-1, 1]$. So the arcsin is defined on the restriction in which this function is strictly increasing and bijective, that is

$$\sin\Big|_{[-\frac{\pi}{2}, \frac{\pi}{2}]} : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \longrightarrow [-1, 1] \implies \arcsin : [-1, 1] \longrightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

The domain is therefore

$$\text{dom } \arcsin = [-1, 1]$$

The sign coincides with the sign of the argument.

Domain:
Recall that $\arcsin(t)$ is defined for $t \in [-1, 1]$, so we must impose this as the domain:

$$-1 \leq \left|x^3 - \frac{1}{2}\right| \leq 1$$

Since the absolute value is always positive we have

$$\left|x^3 - \frac{1}{2}\right| \leq 1 \to -1 \leq x^3 - \frac{1}{2} \leq 1 \to -\frac{1}{\sqrt[3]{2}} \leq x \leq \sqrt[3]{\frac{3}{2}}$$

So we have

$$\text{dom } f = \left[-\frac{1}{\sqrt[3]{2}}, \sqrt[3]{\frac{3}{2}}\right]$$

Here we can notice that since the domain is not symmetric wrt the origin, the function does not have any symmetries, that is: A function can have symmetry **only if** its domain is symmetric with respect to the origin

$$x \in \operatorname{dom} f \implies -x \in \operatorname{dom} f$$

Moreover, since it has a finite domain it cannot be periodic.

Sign:
In our case since we have only positive values, the function will always be positive

$$\arcsin(t) \in \left[0, \frac{\pi}{2}\right] \implies \arcsin(t) \geq 0$$

The root is at $\left|x^3 - \frac{1}{2}\right| = 0 \to x = \sqrt[3]{\frac{1}{2}}$

3) Exponential

Determine domain, sign and possible symmetries and periodicity of the following function:

$$f(x) = \log(e^{2x} - 4e^x + 4)$$

First we do a substitution, $y \overset{\triangle}{=} e^x$ and therefore

$$f(y) = \log(y^2 - 4y + 4)$$

Domain:
From the log we impose $y^2 - 4y + 4 > 0$
The roots are $y_{1,2} = 2$ which for x becomes:

$$\begin{aligned} 2 &= e^x \\ \log 2 &= \log e^x \\ x &= \log 2 \end{aligned}$$

so the domain is just dom $f = \mathbb{R} \setminus \{\log 2\} = (-\infty, \log 2) \cup (\log 2, +\infty)$
Since domain is not symmetric wrt to origin, we don't have any symmetry.

Sign:
Recall $\log(z) \geq 0 \iff z \geq 1$, so

$$y^2 - 4y + 4 \geq 1 \to y^2 - 4y + 3 \geq 0$$

The roots are $y_{1,2} = 1, 3 \to x_{1,2} = 0, \log 3$
The inequality holds for $x \leq 0 \cup x \geq \log 3$

So $f(x) \geq 0 \iff x \leq 0$ or $x \geq \log 3$
And the roots are at $x = 0,\ x = \log 3$

7) Hyperbolic Cosine

Determine domain, sign and possible symmetries and periodicity of the following function:

$$f(x) = \frac{1}{\cosh(\sin(x))}$$

What is the hyperbolic cosine?

$$\cosh(t) = \frac{e^t + e^{-t}}{2}$$

So this is always defined and is always positive.
This is also an even function:

$$\cosh(-t) = \frac{e^{-t} + e^t}{2} = \cosh(t)$$

Domain:
First define $a = \sin(x) \in [-1, 1]$
The denominator cannot be 0 so

$$\cosh(a) = \frac{e^a + e^{-a}}{2} \neq 0$$

Which is valid $\forall a$ as $e^k > 0 \quad \forall k \in \mathbb{R}$.

So

$$\text{dom } f = \mathbb{R}$$

Sign:
Hyperbolic cosine is always positive so also this function is always positive with no zero

Symmetry:
Cosh is even $\cosh(x) = \cosh(-x)$, sin is odd $\sin(-x) = -\sin(x)$.
Try even:

$$f(-x) = \frac{1}{\cosh(\sin(-x))} = \frac{1}{\cosh(-\sin(x))} = \frac{1}{\cosh(\sin(x))} = f(x)$$

So it is even!

Is it periodic?
As before impose

$$f(x + T) = f(x) \quad \text{for all } x \in \mathbb{R}$$

Substituting

$$\frac{1}{\cosh(\sin(x + T))} = \frac{1}{\cosh(\sin(x))} \to \cosh(\sin(x + T)) = \cosh(\sin(x))$$

Easy way: Try period of $\sin(x)$ (2π):

$$\begin{aligned} \cosh(\sin(x + 2\pi)) &= \cosh(\sin(x)) \\ \cosh(\sin(x)) &= \cosh(\sin(x)) \end{aligned}$$

Harder way:
Since cosh has even symmetry:

$$\cosh(t) = \cosh(-t) \implies \cosh(t) = \cosh(|t|) \quad \forall t \in \mathbb{R}$$

And therefore (prove at home):

$$\cosh(|u|) = \cosh(|v|) \iff |u| = |v|$$

And thus impose:

$$|\sin(x + T)| = |\sin(x)| \quad \forall x \in \mathbb{R}$$

Squaring both sides:

$$\sin^2(x + T) = \sin^2(x)$$

Using the identity $\sin^2(\theta) = \frac{1 - \cos(2\theta)}{2}$:

$$\frac{1 - \cos(2x + 2T)}{2} = \frac{1 - \cos(2x)}{2} \iff \cos(2x + 2T) = \cos(2x) \quad \forall x \in \mathbb{R}$$

Therefore the period is the same as $\cos(2x)$ which is $T = \pi$ and thus 2π is not the fundamental period (it is a period nonetheless).

10)

Determine domain, sign and possible symmetries and periodicity of the following function:

$$f(x) = \frac{\log(\sin(x))}{\sin(x) - 1}$$

Domain:
Denominator forces $\sin(x) \neq 1$ which holds for $x \neq \frac{\pi}{2} + 2k\pi \quad k \in \mathbb{Z}$
The log forces $\sin(x) > 0$ which is true for $x \in (2k\pi, \pi + 2k\pi) \quad k \in \mathbb{Z}$
so the generic k-th interval is $(2k\pi, \frac{\pi}{2} + 2k\pi) \cup (\frac{\pi}{2} + 2k\pi, \pi + 2k\pi)$
The domain is the union of all these intervals:

$$\text{dom } f = \bigcup_{k \in \mathbb{Z}} (2k\pi, \frac{\pi}{2} + 2k\pi) \cup (\frac{\pi}{2} + 2k\pi, \pi + 2k\pi)$$

which is not symmetric wrt the origin and thus there is no symmetry

Sign:
The denominator is always negative since $\max \sin(x) = 1$ and thus $\max \sin(x) - 1 = 0$ (which is excluded by domain)
The numerator is also always negative:

$$\log(\sin(x)) \geq 0 \iff \sin(x) \geq 1 \iff \sin(x) = 1$$

which is excluded from the domain

Therefore this function is always positive

Periodicity:
Notice that the domain itself needs to be
periodic (or to be $\mathbb{R}$). In this case the domain is given by intervals given by $\sin(x)$ and it can be shifted only by $2k\pi$ to hold correctly. Therefore we can directly try with $T = 2\pi$:

$$f(x + 2\pi) = \frac{\log(\sin(x + 2\pi))}{\sin(x + 2\pi) - 1} = \frac{\log(\sin(x))}{\sin(x) - 1} = f(x)$$

3) Symbols

- $\Longrightarrow$: implies, sufficient condition ($P \implies Q$ means P is sufficient for Q, Q is necessary for P).
- $\Longleftarrow$: implied by, necessary condition
- $\Longleftrightarrow$: necessary + sufficient, if and only if "iff"
- $\emptyset$: empty set
- $\mathbb{N}$: natural numbers; $\{0, 1, 2, \dots\}$
- $\mathbb{Z}$: integer numbers; $\{\dots, -2, -1, 0, 1, 2, \dots\}$
- $\mathbb{Q}$: rational numbers; $\left\{\frac{m}{n}\ m, n \in \mathbb{Z}, n \neq 0\right\}$
- $\mathbb{R}$: real numbers
- $\mathbb{C}$: complex numbers
- $\cup$: Union; or
- $\cap$: Intersection; and
- $\exists$: there exists